

Structure	Page Nos.
10.0 Introduction	1
10.1 Objectives	1
10.2 Introduction to WLANs	2
10.3 Introduction to WAP	4
10.4 Introduction to WML	7
10.5 Bluetooth	9
10.6 Wi-Max	10
10.7. Zig-Bee and Wi-Fi	11
10.8 Summary	13
10.9 Solutions / Answers	13
10.9 Further Readings	15

10.0 INTRODUCTION

A wireless LAN (WLAN) is a wireless computer network that links two or more devices using wireless communication to form a local area network (LAN) within a limited area such as a home, school, computer laboratory, campus, or office building. This gives users the ability to move around within the area and remain connected to the network. Through a gateway, a WLAN can also provide a connection to the wider Internet.

Wireless LANs based on the IEEE 802.11 standards are the most widely used computer networks in the world. These are commonly called Wi-Fi, which is a trademark belonging to the Wi-Fi Alliance. They are used for home and small office networks that link together laptop computers, printers, smart phones, Web TVs and gaming devices with a wireless router, which links them to the internet. Hotspots provided by routers at restaurants, coffee shops, hotels, libraries, and airports allow consumers to access the internet with portable wireless devices.

It operates in various frequency bands and offers different speeds (802.11n, 802.11ac, 802.11ax) with increasing capabilities.

PAN (Personal Area Network), on the other hand, focuses on connecting devices within a very small range, usually within a few meters. Bluetooth and Zig-Bee are common PAN protocols that facilitate communication between devices like smart phones, tablets, laptops, and smart home gadgets.

In essence, WLAN enables broader wireless connectivity over a larger area, while PAN facilitates short-range connections between nearby devices.

Conceptually, the difference between a PAN and a Wireless Local Area Network (WLAN) is that the former tends to be cantered around one person while the latter is connected without wires and serves multiple users.

10.1 OBJECTIVES

Wireless Local Area Networks (WLANs) have several objectives like wireless connectivity allowing users to move freely while maintaining network connectivity with cost efficiency and scalability. It also convenient and quick access to network resources without physical connection as well as delivering the high data rates with

ease of management and security also availing support for applications over cloud based services.

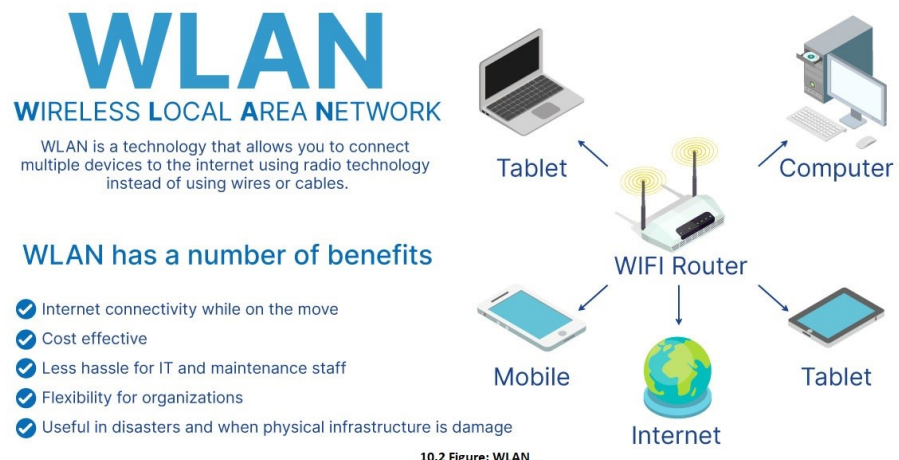
Personal Area Network (PAN) has short range connectivity for convenience allowing data sharing quickly and able to synchronize devices enabling security as well as privacy. PAN network connections can either be wired or wireless. Wired connection methods include USB and FireWire; wireless connection methods include Bluetooth (the most common), Wi-Fi, IrDA, and Zig-Bee.

10.2 INTRODUCTION TO WLANs

The LAN network could not make available the flexibility to move around without being tethered to a physical connection. And this was hampering the suppleness and productivity of the user. In order to handle this problem, researchers came up with WLAN technology.

In simple terms, WLAN or Wireless LAN is a technology that allow devices to connect to the internet wirelessly without need cable because of this feature, WLANs are being used in a variety of places, such as homes, businesses, schools, etc., and have become an vital part of our everyday lives.

WLAN is a technology that allows you to connect multiple devices to the internet using radio technology instead of using wires or cables as shown in the below figure 10.2. Wireless Local Area Network or Wireless LAN allows people to connect devices, such as laptops, mobiles, etc., to the internet without connecting them by cables. WLAN works in a particular area, such as a school, home, campus, or office building.



The Wireless LAN allows you to move around in a specific area and still stay connected to the internet technology that is vital because it makes easier for people to access the internet from almost anywhere thus increasing and boosts the productivity and flexibility of the users.

The researchers started exploring the idea of wireless communication in the 1970s. It was in the early 1990s that the first commercial, WLAN, was launched. The world's first wireless computer communication network, ALOHA net, was developed by a professor at the University of Hawaii named Norman Abramson.

At first, WLAN was unpopular due to its slow speed and high cost. However, with advances in technology, the speed and range of WLAN are improving, and it is

becoming more affordable. Today, it is very common for people to use WLAN in homes, businesses, and public places. WLAN, also known as Local Area Wireless Network (LAWN), transfers data using frequencies such as 2.4 GHz and 5 GHz.

WLANs can be classified into two types: 1. Infrastructure WLAN 2. Ad-hoc WLAN

Let's look at each type in detail.

Infrastructure WLAN uses a central device called a wireless access point (WAP) to connect devices to the network. The WAP is usually connected to a wired network and provides wireless connectivity to devices within its coverage area.

For example, in a home network, the wireless router is the WAP. With an infrastructure WLAN, devices can connect to each other and the internet through the WAP.

Ad-hoc WLAN allows devices to connect each other directly without the need for a central access point and this type of network is useful when devices need to communicate with each other but no existing infrastructure network is available.

For example, if we are in a park with a group of friends, and want to share photos, to complete this, we can create an ad hoc WLAN network to connect your devices. Ad hoc WLAN networks are also commonly used in peer-to-peer file-sharing applications.

To understand the working of WLAN is simple almost the same as the working of LAN networking and the difference is that WLAN can connect your device to the network without a cable.

WLAN uses radio waves to send and receive data between devices, like your laptop and a wireless router. The router acts as a central hub that allows multiple devices to connect to the network and communicate.

For example, let's say that if you want to connect your laptop to a Wireless LAN in your home, first turn on your laptop's wireless network adapter, which detects any available wireless networks within range. Then select your home network from the list of available networks and enter the password to connect. Once it connected, our laptop can send and receive data from the router, allowing us to access the internet and perform various other activities, such as sharing files with other devices on the network, printing documents wirelessly, etc.

Applications of WLAN:

The WLAN network has various applications, depending upon the place it is being used such as at home, it will allow you to connect devices such as laptops, smart phones, and smart TVs to the internet.

In office places, the Wireless LAN network will allow your workers to connect their devices to the internet and share resources such as printers and files with other.

The Wireless LAN network will allow customers to access the internet in public places, such as coffee shops or other also being used in IoT devices such as smart thermostats and security cameras in order to improve flexibility.

Security and Authentication of WLANs can implement various security mechanisms, including encryption and authentication, to ensure secure data communication and prevent unauthorized access.

Overall, WLAN technology has become an essential part of modern connectivity, providing the convenience of wireless communication and enabling the growth of mobile devices, IoT, and the digital transformation of various industries.

Advantages of WLAN:

Here are some of the advantages of Wireless LAN:

Flexibility: Users can move around within the Wireless LAN coverage area while maintaining network connectivity.

Less expensive: Setting up a Wireless LAN network is less expensive in comparison to a wired network. It is so because a Wireless LAN requires fewer physical components, such as cables and switches, thus reducing the cost.

Easy installation: Wireless LAN is easy to install as you don't have to run cables or install ports in walls and ceilings.

Easily scalable: Wireless LAN networks are easy to scale as you need to add more access points to increase coverage.

Disadvantages of WLAN:

Less secure: As wireless signals are more prone to cyber attack, Wireless LAN is less secure if not configured properly than a LAN network.

Limited range: The area covered by a Wireless LAN is not quite large, and in order to increase the Wireless LAN coverage area, you require additional access points.

Bandwidth issues: WLANs may experience bandwidth limitations due to the shared nature of the wireless medium, which can result in slower speeds.

Interference: WLANs are susceptible to interference from other wireless devices and can experience ruined the performance or dropped connections as a result.

Reliability issues: The range and performance of the wireless LAN network can be affected by various factors, such as rain, the number of obstacles, etc. Hence, because of these factors, the chances of dropping the connection increase greatly.

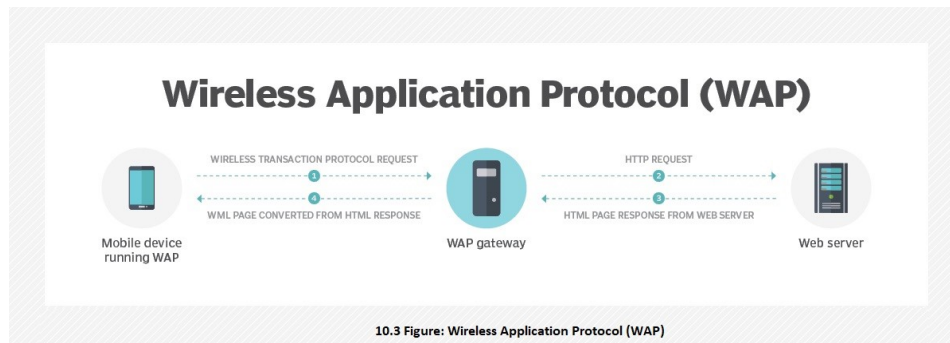
Check Your Progress 1

1. ----- is a technology that allow devices to connect to the internet wirelessly without need cable.
2. WLAN uses -----to send and receive data between devices, like your laptop and a wireless router.
3. Advantages of WLAN?

10.3 INTRODUCTION TO WAP

"WAP" could refer to two different concepts: Wireless Application Protocol or Wireless Access Point, Wireless Application Protocol (WAP) was a standardized technology for accessing information and services on the internet through mobile devices, particularly early mobile phones. It aimed to provide a way to view simplified web content on small screens with limited processing power and bandwidth. WAP used protocols such as Wireless Markup Language (WML) to structure and present content shown below figure 10.3 WAP.

Key points about Wireless Application Protocol are the purpose of WAP was designed to enable mobile devices to access internet-based services and content, including news, weather, email, and more.



A Wireless Access Point (WAP), also known simply as an Access Point (AP), is a networking device that allows wireless devices to connect to a wired network using Wi-Fi or other wireless technologies. Here's what we need to know about WAPs:

Network Connectivity: A WAP bridges the gap between wireless devices and a wired network. It provides a point of access for wireless devices to connect to the larger network, such as a local area network (LAN) or the internet.

Wireless Communication: WAPs use radio waves to transmit and receive data to and from wireless devices like laptops, smart phones, and tablets. They facilitate wireless communication by acting as a relay between the wireless clients and the wired network.

What is Wireless Application Protocol (WAP)?

Wireless Application Protocol (WAP) is a specification for a set of communication protocols to standardize the way wireless devices, such as mobile phones and radio transceivers, can be used for internet access, including email, the web, newsgroups and instant messaging.

WAP was conceived in 1997 by Ericsson, Motorola, Nokia and Unwired Planet (now Enea Openwave Mobility) at an event known as the WAP Forum. While wireless internet access was possible before the introduction of WAP, different manufacturers used varying technologies, and WAP was intended to be an industry standard. However, WAP is now considered obsolete, as modern devices use networks and browsers that function similarly to those on PCs.

How does WAP work?

WAP describes a protocol suite designed to create interoperability between WAP equipment, such as mobile phones that use the protocol, and WAP software, such as WAP-enabled web browsers and network technologies. Before the introduction of WAP, mobile data access capabilities varied widely based on a user's device and mobile provider.

While the WAP protocol was formed to standardize mobile data access, it was also a tool for overcome the device and carrier related limitations that often resulted in mobile users having a dire experience. It did this in several ways:

WAP supported existing and widely used internet standards such as Internet Protocol, User Datagram Packets and XML.

The WAP protocol was optimized for the high-latency, low-bandwidth mobile networks of the time, which were notorious for dropping the connection before a page could fully render.

The Wireless Markup Language (WML) format allowed WebPages to provide without regard for a user's mobile hardware, much like Hypertext Markup Language (HTML) code can be rendered by any browser, apart from of what type of hardware the browser is running on.

Why use WAP?

Introduced in 1999, WAP proposed the following benefits for wireless network operators, content providers and end users:

Wireless network and mobile phone operators. WAP was designed to improve existing wireless data services, such as voicemail, while enabling the development of additional new mobile applications. These applications could be developed without any further infrastructure changes or phone modifications.

Content providers. WAP created a market for additional applications and mobile phone functionalities for third-party application developers to exploit. Writing applications in WML was proposed as a new programming language for developers to create effective mobile device applications.

End users. Mobile phone users would benefit from easy, secure access to online services, such as banking, entertainment, messaging and other information, on mobile devices. WAP could also allow access to intranet information -- such as corporate databases and business applications.

Despite these proposed advantages, WAP did not experience widespread adoption in many countries, and its use declined significantly around 2010 due to extensive HTML compatibility in mobile phones.

WAP model

The WAP model works similarly to the traditional client-server model but depends on an additional component called a *WAP gateway*. This gateway's job is to act as an intermediary between mobile devices and the internet.

Mobile devices of the time lacked the hardware resources commonly found in today's devices. As such, devices were usually equipped with lightweight mobile browsers, also known as *minibrowsers* or *micro browsers*. When a user entered a URL into their device's browser, the request was sent to a WAP gateway. This gateway would visit the site on the device's behalf, retrieve the requested page and then convert the page into WML format. That WML code would then be sent to the device that would render the page.

WAP protocol stack

The WAP standard describes the following protocol stack for interoperability of WAP devices, equipment, software and other technologies, which includes the following: Wireless Application Environment for mobile device specifications and programming languages such as WML; Wireless Session Protocol, which manages connection suspensions and reconnections; Wireless Transaction Protocol, which manages transaction support for requests and responses to servers; Wireless Transport Layer Security for managing privacy, authentication and data integrity through public key

cryptography; and Wireless Datagram Protocol, an adaptation layer for consistent data formats in the other layers that defines how data flows to the sender from the receiver.

Advantages of WAP

The main benefit brought about by WAP was that it made broad internet access possible for mobile devices. Before adopting WAP, mobile carriers typically offered proprietary and extremely limited mobile access. A carrier might offer a few dedicated services providing stock quotes, movie listings, weather, news headlines and sports. However, general web access typically wasn't an option.

In addition to making broad internet access possible, WAP improved access speeds through data compression and helped reduce the number of timeouts and connection failures that had previously plagued mobile access.

Disadvantages of WAP

The primary disadvantage of WAP was that it was never universally adopted. In some areas, mobile providers held back adoption by charging significant additional fees for data access.

Another disadvantage to WAP was that mobile browsers of the time lacked the capabilities of modern browsers. As a result, WAP sometimes failed to render WebPages on mobile devices correctly. Larger and more complex pages often couldn't be rendered at all.

Check Your Progress 2

1. What is Wireless Application Protocol (WAP)?
2. Why use WAP?
3. Disadvantages of WAP

10.4 INTRODUCTION TO WML

As of last knowledge update in September 2021, "WML" could refer to "Wireless Markup Language," which was an older markup language used for creating content meant to be displayed on wireless devices, particularly mobile phones with limited internet capabilities. However, it's important to note that WML is a technology that has largely become obsolete due to advancements in mobile web technologies and the widespread adoption of more capable devices.

Here's a brief overview of Wireless Markup Language (WML):

WML was designed to structure and format content for wireless devices with small screens and limited processing power. It allowed developers to create lightweight and simple web pages tailored for mobile browsing.

WML syntax was similar to HTML, but it was specifically optimized for mobile devices. It included tags for defining decks (pages), cards (screens), text formatting, links, images, and more.

Deck and Card Structure: WML documents were organized into decks and cards. A deck could contain multiple cards, each representing a screen or page within the mobile application.

WML Decks and Cards:

A main difference between HTML and WML is that the basic unit of navigation in HTML is a page, while that in WML is a card. A WML file can contain multiple cards and they form a deck.

When a WML page is accessed from a mobile phone, all the cards in the page are downloaded from the WAP server. So if the user goes to another card of the same deck, the mobile browser does not have to send any requests to the server since the file that contains the deck is already stored in the wireless device.

You can put links, text, images, input fields, option boxes and many other elements in a card.

WML Program Structure:

Following is the basic structure of a WML program;

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.2//EN"
"http://www.wapforum.org/DTD/wml12.dtd">

<wml>

<card id="one" title="First Card">
<p>
This is the first card in the deck
</p>
</card>

<card id="two" title="Second Card">
<p>
Ths is the second card in the deck
</p>
</card>

</wml>
```

10.4 Figure: WML Program Structure

The first line of this text says that this is an XML document and the version is 1.0. The second line selects the document type and gives the URL of the document type definition (DTD).

One WML deck (i.e. page) can have one or more cards as shown above. We will see complete details on WML document structure in subsequent chapter.

Unlike HTML 4.01 Transitional, text cannot be enclosed directly in the <card>...</card> tag pair. So you need to put a content inside <p>...</p> as shown above in 10.4 Figure: WML Program.

Due to the limited capabilities of mobile devices at the time, WML lacked advanced features like JavaScript support and complex layouts. It focused on providing basic text, images, and links.

WML was part of the larger Wireless Application Protocol (WAP), which aimed to provide a standardized framework for wireless internet access. WAP included protocols for communication and content rendering.

Transition to Modern Technologies:As mobile devices evolved and gained more powerful browsers, support for HTML, CSS, and JavaScript became the norm this led to the decline of WML and WAP in favour of more versatile mobile web technologies.

Mobile Responsive Design: In today's landscape, responsive web design using HTML5, CSS3, and JavaScript allows websites to adapt to various screen sizes and devices, rendering the need for WML unnecessary.

Given the advancements in mobile technology and the shift to modern web standards, WML has largely become outdated and is no longer widely used. Developers now focus on creating responsive and adaptive web content that can seamlessly adjust to the capabilities of different devices and screen sizes.

Check Your Progress 3

1. Brief overview of Wireless Markup Language (WML)?
2. Main difference between HTML and WML?

10.5 BLUETOOTH

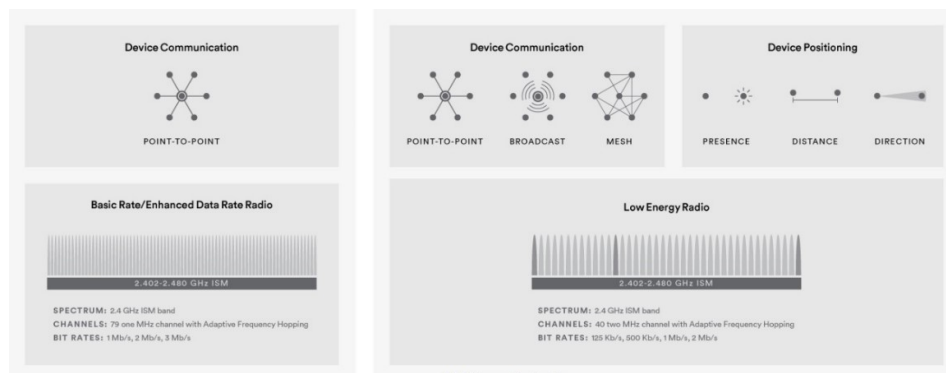
Bluetooth is a wireless communication technology that enables devices to exchange data and information over short distances without the need for cables. It's commonly used for connecting devices like smartphones, laptops, headphones, speakers, and various smart devices. Here are some key points about Bluetooth:

Wireless Connectivity used by Bluetooth which uses radio waves to establish wireless connections between devices which operates in the 2.4 GHz frequency range and short range of Bluetooth connections typically have a range of around 10 meters (approximately 30 feet), making it suitable for personal area networks.

Bluetooth Low Energy (BLE) is a version of Bluetooth designed for energy-efficient communication. It's commonly used in devices with limited battery life, such as fitness trackers and smart watches. Pairing to establish a connection, devices need to be paired and this involves a one-time setup process where the devices exchange security keys to ensure secure communication.

Bluetooth profiles define the features and capabilities that devices support. Common profiles include Headset, Hands-Free, A2DP (Advanced Audio Distribution Profile), and HID (Human Interface Device). Version Evolution: Bluetooth technology has gone through several versions, each offering improvements in terms of data speed, range, and energy efficiency. Some notable versions include Bluetooth 2.0, 3.0, 4.0 (BLE), 4.2, 5.0, and the latest at the time of my last update, Bluetooth 5.1.

Bluetooth is used for a variety of applications, including hands-free calling in cars, wireless audio streaming, connecting wireless headphones and ear buds, file sharing between smart phones, and connecting smart home devices. Bluetooth mesh networking is a feature that allows multiple devices to form a network and communicate with each other, extending the range and creating more complex setups for applications like smart lighting and home automation shown in the below figure 10.5.



10.5 Figure: Bluetooth

Bluetooth incorporates security measures to protect data during transmission. Pairing and encryption help ensure that data is not intercepted by unauthorized devices

Bluetooth can sometimes experience interference from other devices operating in the same frequency range, such as Wi-Fi networks and microwave ovens. Most modern devices are equipped with Bluetooth capabilities, making it a widely compatible technology for wireless communication.

Bluetooth has become a ubiquitous technology for wireless connections, contributing to the growth of the Internet of Things (IoT) and enabling seamless communication between various devices.

Check Your Progress 4

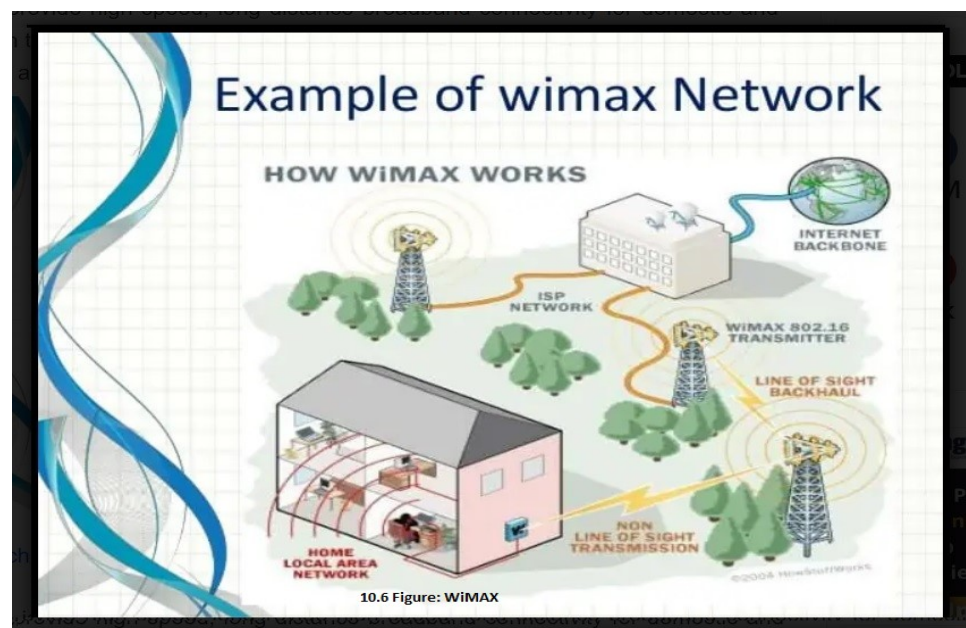
1. What is Bluetooth technology?
2. What is Bluetooth Low Energy (BLE)?

10.6 WiMAX

WiMAX, short for Worldwide Interoperability for Microwave Access, is a wireless communication technology that provides high-speed, long-range broadband connectivity. It was developed to offer an alternative to traditional wired broadband connections and to serve areas with limited or no wired infrastructure.

WiMAX, the Worldwide Interoperability for Microwave Access, is a telecommunications technology aimed at providing wireless data over long distances in a variety of ways, from point-to-point links to full mobile cellular type access. It is the final leg of delivering wireless broadband connectivity from a communications provider to a customer and an alternative to cable and DSL. It is based on a Broadband Wireless Access standard of the Institute of Electrical and Electronics Engineers (IEEE 802.16).

WiMAX and Wi-Fi are complementary and while WiMAX is a broadband connection to the Internet at service quality, Wi-Fi is a wireless local area network and below is the example figure 10.6 for WiMax.



Here are some key aspects of Wi-MAX:

WiMAX is designed to deliver broadband internet access wirelessly, allowing users to connect to the internet without relying on cables or DSL lines. One of the distinguishing features of WiMAX is its ability to cover long distances, making it suitable for providing connectivity to rural or underserved areas where laying cables may not be practical or cost-effective.

WiMAX offers high data transfer rates, allowing for multimedia streaming, online gaming, and other data-intensive activities. WiMAX operates in a point-to-multipoint architecture, meaning that a single base station (access point) can connect with multiple customer premises equipment (CPE) devices.

WiMAX can operate in various frequency bands, including licensed and unlicensed spectrum, depending on regulatory and deployment considerations and initially focused on providing fixed wireless access (similar to a wired broadband connection). However, there were also efforts to develop mobile versions (WiMAX 16e and later WiMAX 16m) that aimed to provide mobile broadband services.

WiMAX supports quality of service (QoS) mechanisms, allowing for prioritization of different types of data traffic to ensure optimal performance for applications like voice, video, and data. WiMAX can serve as a backhaul technology for cellular networks, helping to connect cell towers and base stations to the core network.

Challenges and evolutions while WiMAX technology held promise it faced competition from other wireless technologies, notably 4G LTE (Long-Term Evolution). LTE gained widespread adoption and ultimately became the dominant technology for mobile broadband.

WiMAX is a legacy system that did find some niche applications and deployments, especially in certain regions and industries, but its adoption was limited compared to LTE.

WiMAX 2 (also known as WiMAX 802.16m) was an evolution of the original WiMAX standard that aimed to provide higher data rates and improved performance. However, it didn't achieve widespread adoption.

In summary, WiMAX was developed as a wireless broadband technology with long-range capabilities, but it faced competition from other technologies like LTE. While it didn't become as widely adopted as some other wireless standards, it still had an impact on the evolution of wireless communication and contributed to discussions around extending broadband connectivity to underserved areas.

Check Your Progress 5

1. What is WiMAX?
2. How WiMAX supports?

10.7 ZIG-BEE AND WI-FI

Zig-Bee and Wi-Fi are both wireless communication technologies, but they serve different purposes and are optimized for different types of applications. The purpose of Zig-Bee is designed for low-power, low-data-rate applications, particularly in the context of the Internet of Things (IoT). It focuses on enabling devices to communicate wirelessly over short distances while consuming minimal energy.

It ranges Zig-Bee relatively short, typically around 10 to 100 meters, making it suitable for local area networks and home automation within a confined space. Zig-Bee devices are designed to be energy-efficient, allowing them to operate on battery power for extended periods or even with energy harvesting methods.

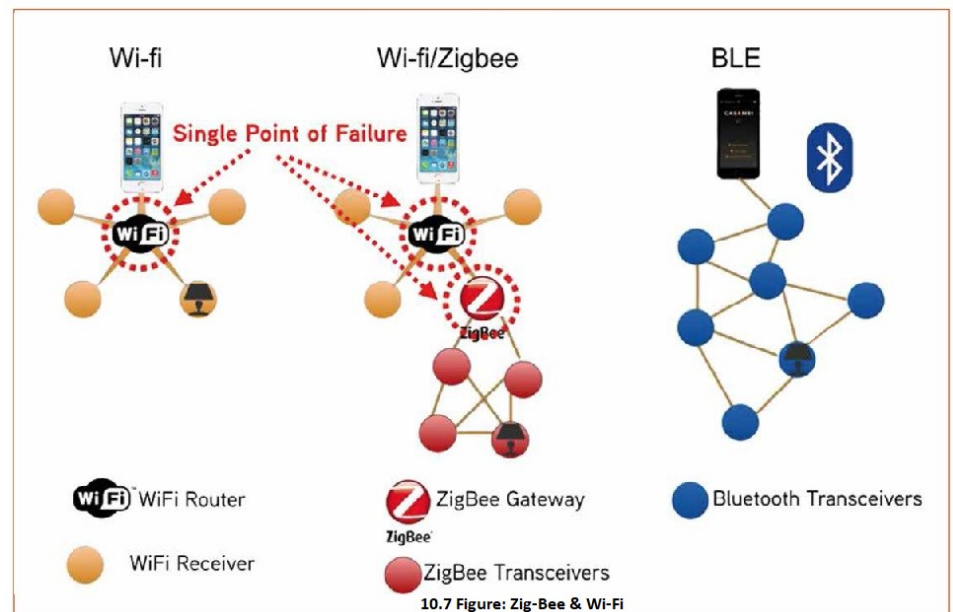
Zig-Bee supports mesh networking, where devices can communicate directly with each other and relay messages for devices out of direct range. This enhances network coverage and reliability.

Data Rate offered by Zig-Bee is lower compared to Wi-Fi, which makes it suitable for applications where transmitting small amounts of data over time is more important than high-speed data transfer.

Applications of Zig-Bee is commonly used in home automation (smart lighting, thermostats, door locks), industrial monitoring and control, healthcare, and sensor networks.

Zig-Bee operates in the 2.4 GHz frequency band, which is shared with other wireless technologies like Wi-Fi and Bluetooth. This can potentially lead to interference in crowded frequency environments.

Wi-Fi is a wireless networking technology that allows devices such as computers (laptops and desktops), mobile devices (smart phones and wearable's), and other equipment (printers and video cameras) to interface with the Internet. It allows these devices--and many more--to exchange information with one another, creating a network. Internet connectivity occurs through a wireless router. When you access Wi-Fi, you are connecting to a wireless router that allows your Wi-Fi-compatible devices to interface with the Internet, below is the figure for 10.7 Zig-Bee and Wi-Fi.



Wi-Fi is designed to provide high-speed wireless internet access over relatively short to medium distances. It offers broad coverage for devices needing fast data transfer rates. Wi-Fi has a range of around 10 to 100 meters, similar to Zig-Bee, but its focus is on providing high data transfer rates for internet-related activities.

Data Rates supported are higher, making it suitable for applications like internet browsing, video streaming, online gaming, and data-intensive tasks and Wi-Fi devices

generally consume more power than Zig-Bee devices, which can affect battery life in portable devices.

Wi-Fi is used for a wide range of applications, including home and business networking, public hotspots, smart homes, mobile devices, and also operates in the 2.4 GHz and 5 GHz frequency bands, which are widely used for various wireless technologies. This can lead to potential interference, especially in crowded urban environments.

In summary, Zig-Bee is tailored for low-power, IoT-centric applications with a focus on conserving energy and enabling reliable communication in limited ranges. Wi-Fi, on the other hand, offers higher data rates and is designed for providing wireless internet access and local networking within a broader coverage area. The choice between Zig-Bee and Wi-Fi depends on the specific requirements of the application, considering factors such as power consumption, data rates, and range.

Check Your Progress 6

1. What is the purpose of Zig-Bee?
2. Explain about Wi-Fi?

10.8 SUMMARY

WLAN, or Wireless Local Area Network, is a wireless communication technology that allows devices to connect to the internet and communicate with each other within a limited geographic area without the need for physical wired connections. Key aspects of WLAN include access points (APs) that serve as communication hubs, adherence to IEEE-defined wireless standards, operation in frequency bands like 2.4 GHz and 5 GHz, security measures like encryption and authentication, SSIDs for network identification, support for roaming, varying range and coverage, widespread use in diverse settings, and the need for management and monitoring. WLANs continue to evolve to offer faster speeds, improved security, and better support for the increasing number of connected devices.

A PAN, or Personal Area Network, is a small-scale network designed for connecting devices within a very limited and typically personal space, often within a range of a few meters to a few tens of meters. Key characteristics of PANs include their small coverage area, low power requirements, and the ability to connect devices like smartphones, tablets, laptops, and wearable technology. PAN technologies commonly used include Bluetooth and Wi-Fi Direct, and they facilitate tasks like file sharing, device synchronization, and peripheral connectivity (e.g., connecting a wireless mouse or keyboard to a computer). PANs are an essential part of our interconnected digital lives, enabling seamless device communication and data transfer in personal environments.

10.9 SOLUTIONS / ANSWERS

Check Your Progress 1

1. Wireless LAN is a technology that allow devices to connect to the internet wirelessly without need cable

2. WLAN uses radio waves to send and receive data between devices, like your laptop and a wireless router.
3. Advantages of WLAN
Flexibility, Less expensive, easy installation and easily scalable

Check Your Progress 2

1. What is Wireless Application Protocol (WAP)?

Wireless Application Protocol (WAP) is a specification for a set of communication protocols to standardize the way wireless devices, such as mobile phones and radio transceivers, can be used for internet access, including email, the web, newsgroups and instant messaging.

2. Why use WAP?

WAP was designed to improve existing wireless data services, such as voicemail, while enabling the development of additional new mobile applications. These applications could be developed without any further infrastructure changes or phone modifications.

3. Disadvantages of WAP

The primary disadvantage of WAP was that it was never universally adopted. In some areas, mobile providers held back adoption by charging significant additional fees for data access.

Check Your Progress 3

1. Brief overview of Wireless Mark up Language (WML)?

WML was designed to structure and format content for wireless devices with small screens and limited processing power. It allowed developers to create lightweight and simple web pages tailored for mobile browsing.

2. Main difference between HTML and WML?

A main difference between HTML and WML is that the basic unit of navigation in HTML is a page, while that in WML is a card. A WML file can contain multiple cards and they form a deck.

Check Your Progress 4

1. What is Bluetooth technology?

Bluetooth is a wireless communication technology that enables devices to exchange data and information over short distances without the need for cables. It's commonly used for connecting devices like smartphones, laptops, headphones, speakers, and various smart devices.

2. What is Bluetooth Low Energy (BLE)?

Bluetooth Low Energy (BLE) is a version of Bluetooth designed for energy-efficient communication. It's commonly used in devices with limited battery life, such as fitness trackers and smart watches. Pairing to establish a connection, devices need to be paired and this involves a one-time setup process where the devices exchange security keys to ensure secure communication.

Check Your Progress 5

1. What is WiMAX?

WiMAX, short for Worldwide Interoperability for Microwave Access, is a wireless communication technology that provides high-speed, long-range broadband connectivity. It was developed to offer an alternative to traditional wired broadband connections and to serve areas with limited or no wired infrastructure.

2. How WiMAX supports?

WiMAX supports quality of service (QoS) mechanisms, allowing for prioritization of different types of data traffic to ensure optimal performance for applications like voice, video, and data. WiMAX can serve as a backhaul technology for cellular networks, helping to connect cell towers and base stations to the core network.

Check Your Progress 6

1. What is the purpose of Zig-Bee?

Zig-Bee and Wi-Fi are both wireless communication technologies, but they serve different purposes and are optimized for different types of applications. The purpose of Zig-Bee is designed for low-power, low-data-rate applications, particularly in the context of the Internet of Things (IoT). It focuses on enabling devices to communicate wirelessly over short distances while consuming minimal energy.

2. Explain about Wi-Fi?

Wi-Fi is designed to provide high-speed wireless internet access over relatively short to medium distances. It offers broad coverage for devices needing fast data transfer rates. Wi-Fi has a range of around 10 to 100 meters, similar to Zig-Bee, but its focus is on providing high data transfer rates for internet-related activities.

10.10 FURTHER READINGS

Wireless Local Area Network (WLAN):

1. "Wi-Fi Technology Explained" - Article by Lifewire providing a comprehensive overview of Wi-Fi technology: [Wi-Fi Technology Explained](<https://www.lifewire.com/wi-fi-technology-818262>)
2. "Introduction to Wireless Local Area Networks" - An educational resource by Cisco covering WLAN basics: [Introduction to WLANs](https://www.cisco.com/c/en/us/products/collateral/wireless/aironet-1250-series/design_guide_c07-693245.html)

Personal Area Network (PAN):

1. "Understanding Personal Area Networks (PANs)" - An article by Lifewire explaining PANs and their applications: [Understanding PANs](<https://www.lifewire.com/personal-area-network-pan-818274>)
2. "Bluetooth Technology Overview" - A guide by Bluetooth SIG that provides an in-depth look at Bluetooth technology: [Bluetooth Technology Overview](<https://www.bluetooth.com/learn-about-bluetooth/bluetooth-technology/bluetooth/>)

Wireless Networking and WLAN:

1. "802.11 Wireless Networks: The Definitive Guide" by Matthew S. Gast

2. "Wireless Communications: Principles and Practice" by Theodore S. Rappaport
3. "Wireless Networking Absolute Beginner's Guide" by Michael Miller
4. "CWNA Certified Wireless Network Administrator Official Study Guide" by David D. Coleman and David A. Westcott

****Networking and Communication:****

1. "Computer Networking: Principles, Protocols and Practice" by Olivier Bonaventure (Available online for free: [Computer Networking](<https://book.systemsapproach.org/>))
2. "Networking All-in-One For Dummies" by Doug Lowe
3. "Data Communications and Networking" by Behrouz A. Forouzan

****Bluetooth and PAN:****

1. "Bluetooth Essentials for Programmers" by Albert S. Huang, Larry Rudolph, and Brent A. Miller
2. "Zigbee Wireless Networking" by Drew Gislason